Real-time Detailed Video Analysis of Fruit Flies

UE24CS352A Machine Learning | Mini-Project Write-up

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99.3%
male / female identified correctly

0.64°
average heading error (0-360°)

2.0°
average wing-angle error (0-90°)

52 FPS
analysis speed, 1.7x the 30 FPS video

1 Problem statement

Fruit-fly behaviour, such as courtship, is studied by filming flies and analysing the video. Existing tools (for example FlyTracker) run roughly ten times slower than real time, which makes long recordings slow to process and rules out closed-loop experiments that must react while the fly is behaving.

Goal. Analyse a grayscale video of one male and one female fly and, for **every frame**, output each fly's **sex, position and heading (0-360°)** and the male's **right and left wing angles (0-90°)**, faster than the 30 FPS video, on an ordinary CPU, with classical machine learning (re-implementing S. Herbst, Stanford CS229, 2018).

2 Dataset details

The lab video of the reference project is not public, so we generated our own data with a simulator (`src/synth.py`) that draws a smaller male and a larger, striped female in a circular well, with wings at any angle, legs, blur, noise and debris. Because the images are drawn, every label is exact.

Item	Details
Labelled frames	900 grayscale images (512 × 512): 78% flies apart, 10% close, 12% touching
Train / test split	600 training / 300 test frames (2/3 : 1/3), split by frame
Labels	per fly: sex, position, heading 0-360°; for the male also right and left wing angle
Test video	separate 600-frame video (20 s at 30 FPS), different random seed, never used for training

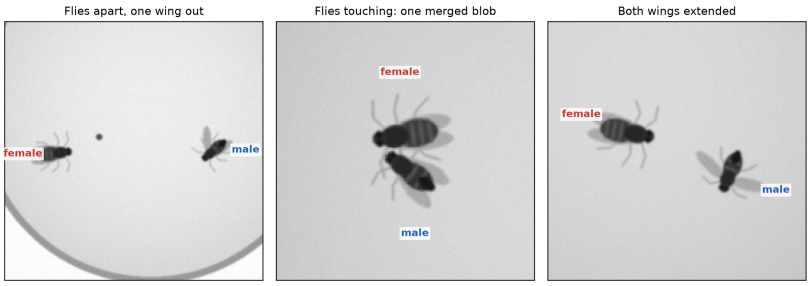


Fig. 1 Simulated examples. When the flies touch they form one merged blob (middle), which the pipeline detects and skips.

3 Approach

Dark pixels are thresholded to find the fly bodies; four small models then work on those regions only, which keeps the pipeline fast (Fig. 2). **HOG** (histogram of oriented gradients) describes a shape by its edge directions and **PCA** compresses it (3,780 / 3,564 numbers to 15 / 40).

Stage	Model	What it does
1 Fly counting	Decision tree	blob area: debris, one fly, or two touching
2 Sex identification	Logistic regression	area + shape of both flies: which is the male
3 Orientation	HOG + PCA + logistic regression	moments give the axis, classifier the head end
4 Wing angles	HOG + PCA + linear regression	upright male, legs masked: right/left wing angle

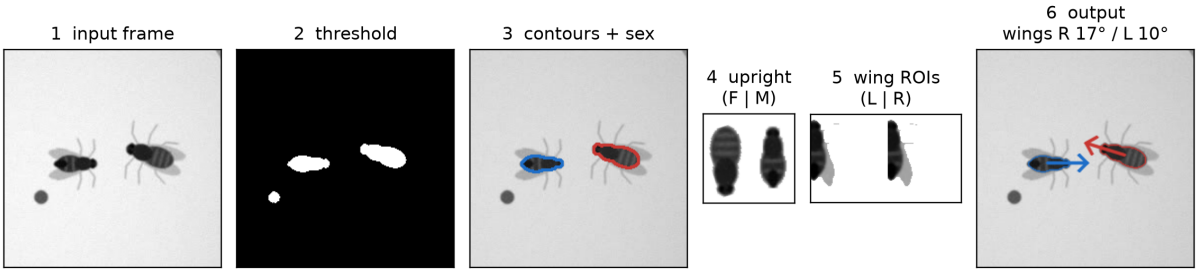


Fig. 2 The pipeline on one frame (blue = male, red = female): threshold, contours, upright patches, wing regions, final output.

4 Implementation overview

- **Python:** OpenCV (thresholding, contours, moments), scikit-image (HOG), scikit-learn (all models, PCA) and Tkinter (GUI).
- `generate_dataset.py` > `train_evaluate.py` > `demo.py` / `gui.py`: create the data, train and test the four models, then analyse a video with a plain-English report or a live window. The whole pipeline is `FlyPipeline.process(frame)` in `src/pipeline.py`.
- Submitted as a private GitHub repository; the README explains setup and the run order.

5 Results

Each model was first tested on the 300 frames it never saw in training; the full pipeline was then run on the held-out 600-frame video (Table 1, Fig. 3).

Table 1. Error on unseen test frames and on training frames (lower is better).

Stage	Test error	Train error
1 Fly counting	0.92%	0.36%
2 Sex identification	0.69%	0.00%
3 Orientation, male	0.00%	0.00%
3 Orientation, female	0.00%	0.00%
4 Wing angles*	5.19°	4.29°

* spread (standard deviation) of the wing-angle error; the true angles vary with a spread of 25° (R^2 = 0.96).

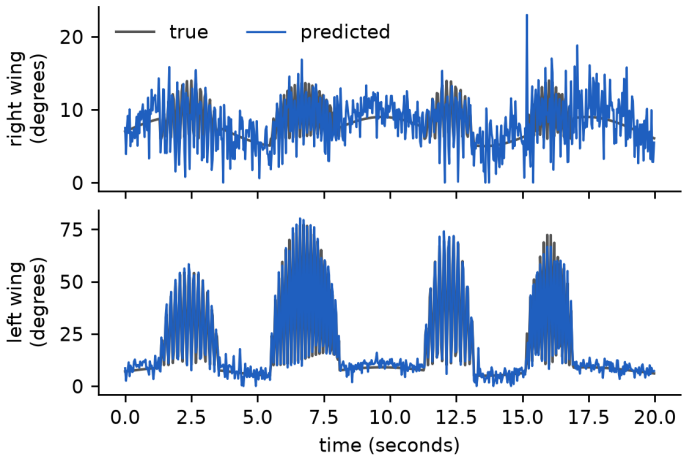


Fig. 3 Predicted vs true wing angle on the test video.

On the test video (600 frames, 1200 flies checked) no frame had to be skipped and the heading was within 15° for 99.8% of flies. The pipeline ran at **52 frames per second** (19.3 ms per frame), 1.7x faster than the video, on a CPU; orientation and wing angles take about 80% of the time. (Speed was measured in our test environment and differs between computers.)

6 Conclusions

- Four small classical models recover sex, orientation and wing angles at **1.7x real time** on a CPU.
- Image moments reduce orientation to a yes/no question (which end is the head), solved with 0% test error here; wing-angle error falls as more PCA components are used (11.7° spread with 1, 5.2° with 40).
- **Limitations:** the data are simulated, so real footage will be harder; touching flies form one blob and are skipped. **Future work:** split touching flies, train on real annotated video.

References [1] S. Herbst, *Real-time Detailed Video Analysis of Fruit Flies*, Stanford CS229 project, 2018. [2] N. Dalal, B. Triggs, Histograms of Oriented Gradients for Human Detection, CVPR 2005. [3] F. Pedregosa et al., Scikit-learn, JMLR 12, 2011. [4] G. Bradski, The OpenCV Library, 2000.